Term 3 — Lesson 1 Practice Activity 3 (C1–C3) with Solutions

Contents

Evaluated criteria

- C1: Identify whether the context uses a population proportion or a population mean and map the symbols correctly in context.
- C2: Write the null and alternative hypotheses in correct notation, keeping the null as "parameter equals benchmark."
- C3: Classify the test direction (left-tailed, right-tailed, or two-tailed) and justify it from the wording of the claim.

1 Problem 1 — Academic routine in a subject class

Create a hypothesis test about a measurable pattern in one of your Grade 11 subjects (for example, completion rates, quiz performance, or time spent on a recurring class routine).

(C1) For your chosen context, identify whether it is a proportion setting or a mean setting; define the parameter in context; explain how sample information would be collected; and state a plausible sample result.

(C2) State the benchmark claim in context; justify the benchmark narratively; state a significance level; then write the null and alternative hypotheses with correct notation, keeping the null as "parameter equals benchmark."

(C3) Classify the test as left-tailed, right-tailed, or two-tailed, and justify the classification from the wording and intention of the claim.

C1

One possible strong response is to study the percentage of students in mathematics class who complete the weekly practice set before Friday. This is a proportion setting. Let p be the true proportion of Grade 11 mathematics students who submit the weekly practice set before Friday. A reasonable sample method is to check one randomly selected week from each of 8 weeks and record all students enrolled in those sections. A plausible sample result is $n = 160$ students observed with $x = 126$ on-time submissions, so $\hat{p} = 126/160 = 0.7875$.

C2

One possible benchmark claim is that at least 75% of students complete the weekly set before Friday, based on the class expectation set at the start of term. A significance level of $\alpha = 0.05$ is reasonable for a school-based decision.

$$H_0\colon p = 0.75, \qquad H_a\colon p > 0.75.$$

This keeps the null as "parameter equals benchmark" and tests whether current performance is above the target.

C3

This is a right-tailed test, because the wording and intention are to check whether the true completion proportion is *higher* than 75%.

2 Problem 2 — School service or shared school-space feature

Create a hypothesis test about a measurable feature of a school service or shared space (for example, library use, cafeteria flow, or computer-lab access).

(C1) For your chosen context, identify whether it is a proportion setting or a mean setting; define the parameter in context; explain how sample information would be collected; and state a plausible sample result.

(C2) State the benchmark claim in context; justify the benchmark narratively; state a significance level; then write the null and alternative hypotheses with correct notation, keeping the null as "parameter equals benchmark."

(C3) Classify the test as left-tailed, right-tailed, or two-tailed, and justify the classification from the wording and intention of the claim.

C1

One possible strong response is to investigate average waiting time for a computer station in the school lab during lunch break. This is a mean setting. Let μ be the true mean waiting time (in minutes) for students to get a free computer station at lunch. A plausible sample method is to record waiting times for a random set of 50 students across several days. A plausible sample result is $n = 50$ and $\bar{x} = 6.4$ minutes.

C2

A benchmark claim could be that the mean waiting time is 7 minutes, based on prior monitoring by the technology coordinator. A significance level of $\alpha = 0.10$ can be justified if the school wants to detect improvements with moderate evidence.

$$H_0\colon \mu = 7, \qquad H_a\colon \mu < 7.$$

The null keeps the benchmark value, while the alternative represents the claim that access has improved (shorter waits).

C3

This is a left-tailed test, because the intended claim is that the true mean waiting time is *less than* the benchmark.

3 Problem 3 — Family or household routine

Create a hypothesis test about a measurable household routine (for example, shared chores, meal timing, or daily technology habits at home).

(C1) For your chosen context, identify whether it is a proportion setting or a mean setting; define the parameter in context; explain how sample information would be collected; and state a plausible sample result.

(C2) State the benchmark claim in context; justify the benchmark narratively; state a significance level; then write the null and alternative hypotheses with correct notation, keeping the null as "parameter equals benchmark."

(C3) Classify the test as left-tailed, right-tailed, or two-tailed, and justify the classification from the wording and intention of the claim.

C1

One possible strong response is to model the proportion of weekday dinners in a household that begin within 15 minutes of the planned time. This is a proportion setting. Let p be the true proportion of weekday dinners that begin within that 15-minute window. A practical sampling method is to track dinner start times over 30 weekdays in one term. A plausible sample result is $n = 30$ weekdays with $x = 21$ on-schedule starts, so $\hat{p} = 0.70$.

C2

A benchmark claim might be that only 60% of dinners start within 15 minutes of plan, based on family experience before introducing a weekly schedule board. Choose $\alpha = 0.05$.

$$H_0\colon p = 0.60, \qquad H_a\colon p > 0.60.$$

The hypotheses test whether the routine has become more consistent than the old benchmark.

C3

This is a right-tailed test, because the claim is that the true proportion is *greater than* 0.60.

4 Problem 4 — Friendship or peer-group setting

Create a hypothesis test about a measurable feature of a peer-group habit (for example, punctuality, response time, or planning consistency for group activities).

(C1) For your chosen context, identify whether it is a proportion setting or a mean setting; define the parameter in context; explain how sample information would be collected; and state a plausible sample result.

(C2) State the benchmark claim in context; justify the benchmark narratively; state a significance level; then write the null and alternative hypotheses with correct notation, keeping the null as "parameter equals benchmark."

(C3) Classify the test as left-tailed, right-tailed, or two-tailed, and justify the classification from the wording and intention of the claim.

C1

One possible strong response is to examine average delay (in minutes) between the agreed meetup time and actual arrival time among a friendship group. This is a mean setting. Let μ be the true mean delay for that group when meeting after school. A sample could be collected by recording delays for 24 meetups chosen over two months. A plausible sample result is $n = 24$ and $\bar{x} = 4.8$ minutes.

C2

A benchmark claim could be that the group's mean delay is 5 minutes, based on past records kept by one member. Using $\alpha = 0.05$, a student may test whether the delay has changed after introducing reminder messages.

$$H_0\colon \mu = 5, \qquad H_a\colon \mu \neq 5.$$

The two-sided alternative matches a "changed from" interpretation rather than a single direction.

C3

This is a two-tailed test, because the intention is to detect whether the true mean delay is *different from* 5 minutes (either higher or lower).

5 Problem 5 — Local business or nearby service

Create a hypothesis test about a measurable feature of a nearby service you can realistically observe (for example, queue time at a shop, order accuracy, or service speed).

(C1) For your chosen context, identify whether it is a proportion setting or a mean setting; define the parameter in context; explain how sample information would be collected; and state a plausible sample result.

(C2) State the benchmark claim in context; justify the benchmark narratively; state a significance level; then write the null and alternative hypotheses with correct notation, keeping the null as "parameter equals benchmark."

(C3) Classify the test as left-tailed, right-tailed, or two-tailed, and justify the classification from the wording and intention of the claim.

C1

One possible strong response is to evaluate the proportion of orders prepared correctly at a neighborhood bakery during afternoon rush. This is a proportion setting. Let p be the true proportion of customer orders that are correct on first delivery in that time period. A plausible sample method is to observe 100 orders across five afternoons. A plausible sample result is $n = 100$, $x = 91$, so $\hat{p} = 0.91$.

C2

A benchmark claim might be that 95% of orders are correct, based on the store's posted quality target. Choosing $\alpha = 0.05$, a student can test whether actual quality is below that standard.

$$H_0 \colon p = 0.95, \qquad H_a \colon p < 0.95.$$

The null fixes the benchmark exactly, and the alternative reflects concern about underperformance.

C3

This is a left-tailed test, because the intended claim is that the true correctness proportion is *lower than* 0.95.

6 Problem 6 — City, transport, environment, or public-space setting

Create a hypothesis test about a measurable feature of a city or public-space situation that Grade 11 students can investigate responsibly (for example, bus arrival timing, park cleanliness indicators, or noise levels near a public area).

(C1) For your chosen context, identify whether it is a proportion setting or a mean setting; define the parameter in context; explain how sample information would be collected; and state a plausible sample result.

(C2) State the benchmark claim in context; justify the benchmark narratively; state a significance level; then write the null and alternative hypotheses with correct notation, keeping the null as "parameter equals benchmark."

(C3) Classify the test as left-tailed, right-tailed, or two-tailed, and justify the classification from the wording and intention of the claim.

C1

One possible strong response is to study city-bus punctuality at a commonly used stop by measuring the proportion of buses arriving no more than 3 minutes late. This is a proportion setting. Let p be the true proportion of buses on that route that meet this punctuality rule during weekday mornings. A plausible sample plan is to record 80 bus arrivals over two weeks. A plausible sample result is $n = 80$, $x = 52$, so $\hat{p} = 0.65$.

C2

A benchmark claim could be that 70% of buses satisfy the 3-minute rule, based on the transit agency's service standard. Using $\alpha = 0.05$:

$$H_0 \colon p = 0.70, \qquad H_a \colon p \neq 0.70.$$

This setup is appropriate when the investigation asks whether real punctuality is different from the published standard, not necessarily only better or worse.

C3

This is a two-tailed test, because the claim investigates whether the true punctuality proportion is *different from* 0.70 in either direction.